

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

M.Tech. (CAD/CAM)

Sem -II University Examination, 2010-11

ME – 709 Advance Engineering Optimization Techniques

Date: 04.01.2011, Tuesday

Time: 10:30 a.m. to 01:30 p.m.

Maximum Marks: 70

Instructions:

1. Figures to the right indicate **full** marks.
2. Make suitable assumptions and draw neat figures wherever if required.

SECTION- I

- Q1 (a)** What is difference between elimination and interpolation methods? **03**
- (b)** State the necessary and sufficient conditions for the minimum of a function $f(x)$. **04**
- (c)** Prove that convex function is unimodal **05**
- Q2 (a)** Define Saddle point with examples **03**
- (b)** Solve using Fibonacci Method **08**
- Maximise $y = 3 + 6x - 4x^2$ in the interval $[0,1]$ for $\epsilon = 0.05$, $n = 6$
- OR**
- Q2 (a)** State the Khun tucker Conditons **05**
- (b)** Write an algorithm for Cubic Search Method for finding minimum value of any function. **06**
- Q3** Attempt Short notes on the following (Any two) **12**
- (a)** C Program on Golden Section Method
- (b)** C Program on Newton Raphson's Method
- (c)** C Program on Bisection Method

SECTION - II

- Q4 (a)** Find out the best work-tool combinations in terms of machinability for the data given in Table 1 using TOPSIS methods. The attributes considered are tool wear rate (TW), specific energy consumed (SE), and processed surface roughness (SR). The weights for the attributes are $WTW = 0.7306$; $WSE = 0.1884$; $WSR = 0.0810$ resp.

12

Table:1 Objective data of the machinability attributes

Work-tool combination	Tool wear rate (m/min)	Specific energy consumed (N)	Surface roughness (μm)
1	0.061	219.74	5.8
2	0.028	761.46	5.8
3	0.034	1,593.48	5.8
4	0.013	2,849.15	6.2

1: TiAl6V4-P20; 2: TiMo32-P20; 3: TiAl5Fe2.5-P20; 4: TiAl6V4-P20 (TiN);

- Q5 (a)** Explain the optimality criteria of a multi variable function 06

- (b)** Consider the function $f(x) = 2x_1^2 + 4x_1x_2^2 - 10x_1x_2 + x_2^2$ and the point $\bar{x} = [0, 0]^T$. Classify $\bar{x}$. 05

OR

- Q5 (a)** Discuss the difference between Genetic Algorithm and Neuro Fuzzy Algorithm. Which one would you rate better in terms of solving optimization problems? 05

- (b)** Solve the function $f(x) = x^2 + 54/x$ using Genetic Algorithm. Find the value of x^* . 06

- Q6** Attempt Short notes on the following (Any two) 12

- (a) Ant Colony Optimisation
- (b) Particle Swarm Optimisation
- (c) Simulated Annealing